
Statin Recommendation Project

Background

- Based around Pooled Cohort Equations
 - Risk assessment tool for calculating 10-year risk of atherosclerotic cardiovascular disease
 - Incorporated into clinical guidelines on cholesterol, hypertension, and aspirin therapy
 - Widely used in clinical practice by primary care providers and cardiologists
- Guidelines recommend the tool as part of the statin therapy recommendation decision making process
 - $\geq 7.5\%$ for above 40
 - $\geq 10\%$ for above 40 and stage 1 hypertension

Background

- Want to look at model robustness across range of values
- Goal to determine if there are certain combinations of non-modifiable factors (age, race, and sex) where the risk calculator always returns intermediate/high risk, regardless of other modifiable clinical or behavioral values

Background

Current Age ⓘ *

Age must be between 20-79

Sex *

Male	Female
------	--------

Race *

White	African American	Other
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Systolic Blood Pressure (mm Hg) *

Value must be between 90-200

Diastolic Blood Pressure (mm Hg) *

Value must be between 60-130

Total Cholesterol (mg/dL) *

Value must be between 130 - 320

HDL Cholesterol (mg/dL) *

Value must be between 20 - 100

LDL Cholesterol (mg/dL) ⓘ ○

Value must be between 30-300

History of Diabetes? *

Yes	No
-----	----

Smoker? ⓘ *

Current ⓘ	Former ⓘ	Never ⓘ
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On Hypertension Treatment? *

Yes	No
-----	----

On a Statin? ⓘ ○

Yes	No
-----	----

On Aspirin Therapy? ⓘ ○

Yes	No
-----	----

Data Source

- National Health and Nutrition Examination Survey (NHANES)
 - Run by National Center for Health Statistics, part of the CDC
 - Combines data from interviews and physical examinations
 - Interviews ⇒ demographic, socioeconomic, dietary, and health related questions
 - Physical exams ⇒ lab tests, medical/dental/physiological measurements
 - Collects nationally representative sample data from 5,000 people per year
 - Dataset published every other year
 - Used datasets from 2011-2012, 2013-2014, 2015-2016, and 2017-2018 surveys

Data Definitions

- Age
 - Age at screening
- High Cholesterol
 - Direct HDL cholesterol (mg/dL)
- Total Cholesterol
 - Total cholesterol (mg/dL)
- Has Hypertension Medication
 - Taking prescription medication for cholesterol
- Sex
 - Male/Female
- Ethnicity
 - Race and hispanic origin
- Diabetes Status
 - Told by doctor that they have diabetes
- Smoking Status
 - Do they currently smoke cigarettes
- SBP
 - Mean reading (mmHg)
- DBP
 - Mean reading (mmHg)
- High Blood Pressure Status
 - Told by doctor that they have high blood pressure

Data Definitions

	SEQN	age	high_chol	total_chol	has_hype_med	sex	ethnicity	is_diabetic	does_smoke	sbp	dbp	has_hbp
0	62164.0	44.0	28.0	190.0	0.0	Female	Non-Hispanic White	0.0	0.0	118.000000	60.000000	0.0
1	62172.0	43.0	73.0	169.0	0.0	Female	Non-Hispanic Black	0.0	1.0	102.000000	71.333333	0.0
2	62177.0	51.0	43.0	198.0	0.0	Male	Non-Hispanic Asian	0.0	1.0	146.666667	70.000000	0.0
3	62179.0	55.0	54.0	206.0	0.0	Male	Non-Hispanic Asian	0.0	0.0	124.666667	76.000000	0.0
4	62191.0	70.0	35.0	118.0	0.0	Male	Non-Hispanic Black	0.0	0.0	158.666667	62.666667	0.0

Data Definitions - Additional Variables

- Ethnicity Binary
 - Black or Other
- Risk Score
 - Calculated using ASCVD risk calculator equations
- Group
 - Other/Black Male/Female
- All Flag (g1_is_rec)
 - All people
 - Risk score is/isn't $\geq 7.5\%$
- HBP Flag (g2_is_rec)
 - People with high blood pressure
 - Risk score is/isn't $\geq 10\%$

⇒ percentages defined by ASCVD, indicates recommendation of high-intensity statin therapy

	SEQN	ethnicity_binary	group	risk_score	g1_is_rec	g2_is_rec
0	62164	Other	Other Female	0.017973	0	0
1	62172	Black	Black Female	0.002583	0	0
2	62177	Other	Other Male	0.117270	1	1
3	62179	Other	Other Male	0.051719	0	0
4	62191	Black	Black Male	0.161240	1	1

Dataset Generation

```
LIBNAME nhanes "/home/u63563888/schiltz24/nhanes/nhanesData/2011_raw";
```

```
/* create tables for each NHanes dataset */
```

```
PROC SQL;
```

```
    CREATE TABLE blood_pressure AS
```

```
    SELECT SEQN, BPQ040A, BPQ090D, BPQ100D, BPD035
```

```
    FROM nhanes.bpq_g;
```

```
QUIT;
```

```
/* join tables */
```

```
PROC SQL;
```

```
    CREATE TABLE nhanes_joined AS
```

```
    SELECT *
```

```
    FROM blood_pressure bp
```

```
        INNER JOIN demographics demo ON bp.SEQN = demo.SEQN
```

```
        INNER JOIN diabetes db ON demo.SEQN = db.SEQN
```

```
        INNER JOIN high_chol hc ON db.SEQN = hc.SEQN
```

```
        INNER JOIN smoking smk ON hc.SEQN = smk.SEQN
```

```
        INNER JOIN total_chol tc ON smk.SEQN = tc.SEQN
```

```
        INNER JOIN bmi ON tc.SEQN = bmi.SEQN
```

```
        INNER JOIN creatinine ctn ON bmi.SEQN = ctn.SEQN
```

```
        INNER JOIN blood_measures bm ON ctn.SEQN = bm.SEQN
```

```
    WHERE RIDAGEYR >= 40 AND RIDAGEYR <= 79
```

```
    ;
```

```
QUIT;
```

Dataset Generation

```
/* clean data */  
DATA nhanes;  
    SET nhanes_joined;  
  
/* include those who were never told they had hypertension */  
IF BPD035 >= 5 AND BPD035 <=80 THEN DO;  
    IF BPQ040A = 1 THEN has_hype_med = 1;  
    ELSE IF BPQ040A = 2 THEN has_hype_med = 0;  
    ELSE has_hype_med = .;  
END;  
ELSE IF BPD035 = . THEN has_hype_med = 0;  
ELSE has_hype_med = .;
```

```
/* get complete cases */  
PROC SQL;  
    CREATE TABLE comp_nhanes AS  
    SELECT *  
    FROM nhanes  
    WHERE seqn IS NOT NULL  
        AND age IS NOT NULL  
        AND high_chol IS NOT NULL  
        AND total_chol IS NOT NULL  
        AND has_hype_med IS NOT NULL  
        AND sex IS NOT NULL  
        AND ethnicity IS NOT NULL  
        AND is_diabetic IS NOT NULL  
        AND does_smoke IS NOT NULL  
        AND sbp IS NOT NULL  
        AND dbp IS NOT NULL  
        AND has_hbp IS NOT NULL  
  
    ;  
QUIT;
```

Dataset Generation

```
PROC SQL;  
    CREATE TABLE nhanes_concat AS  
    SELECT * FROM nh_path.nhanes_cleaned_11  
    UNION  
    SELECT * FROM nh_path.nhanes_cleaned_13  
    UNION  
    SELECT * FROM nh_path.nhanes_cleaned_15  
    UNION  
    SELECT * FROM nh_path.nhanes_cleaned_17  
    ;  
QUIT;
```

```
PROC SQL;  
    CREATE TABLE nhanes_b AS  
    SELECT *,  
    CASE WHEN ethnicity = "Non-Hispanic Black" THEN "Black"  
    ELSE "Other"  
    END AS ethnicity_binary  
    FROM nhanes_concat  
    ORDER BY seqn  
    ;  
QUIT;
```

Dataset Generation

Total rows: 13338 Total columns: 14

Rows 1-100

	SEQN	age	high_chol	total_chol	has_hype_med	sex	ethnicity
1	62164	44	28	190	0	Female	Non-Hispani
2	62172	43	73	169	0	Female	Non-Hispani
3	62177	51	43	198	0	Male	Non-Hispani
4	62179	55	54	206	0	Male	Non-Hispani
5	62191	70	35	118	0	Male	Non-Hispani
6	62199	57	35	171	0	Male	Non-Hispani
7	62200	42	36	223	0	Male	Non-Hispani
8	62209	62	31	174	0	Female	Mexican Am
9	62215	65	52	210	0	Female	Non-Hispani
10	62217	77	46	173	1	Female	Mexican Am
11	62221	41	48	240	1	Female	Non-Hispani
12	62223	54	57	242	0	Male	Non-Hispani
13	62228	50	33	183	1	Male	Non-Hispani
14	62230	75	36	152	1	Male	Non-Hispani
15	62231	48	59	182	0	Female	Mexican Am
16	62232	42	38	239	1	Female	Non-Hispani
17	62236	61	39	205	0	Male	Non-Hispani
18	62237	58	53	195	1	Female	Non-Hispani
19	62248	65	30	219	1	Male	Non-Hispani
20	62251	51	50	232	0	Female	Non-Hispani

Dataset Import

```
# convert variables of type byte string to string
```

```
data['sex'] = data['sex'].str.decode('utf-8')
```

```
data['ethnicity'] = data['ethnicity'].str.decode('utf-8')
```

```
data['ethnicity_binary'] = data['ethnicity_binary'].str.decode('utf-8')
```

```
data.head()
```

	SEQN	age	high_chol	total_chol	has_hype_med	sex	ethnicity	is_diabetic	does_smoke	sbp	dbp	has_hbp	ethnicity_binary
0	62164.0	44.0	28.0	190.0	0.0	Female	Non-Hispanic White	0.0	0.0	118.000000	60.000000	0.0	Other
1	62172.0	43.0	73.0	169.0	0.0	Female	Non-Hispanic Black	0.0	1.0	102.000000	71.333333	0.0	Black
2	62177.0	51.0	43.0	198.0	0.0	Male	Non-Hispanic Asian	0.0	1.0	146.666667	70.000000	0.0	Other
3	62179.0	55.0	54.0	206.0	0.0	Male	Non-Hispanic Asian	0.0	0.0	124.666667	76.000000	0.0	Other
4	62191.0	70.0	35.0	118.0	0.0	Male	Non-Hispanic Black	0.0	0.0	158.666667	62.666667	0.0	Black

Risk Score Calculation

```
risk_scores = []
group_one_flags = []
group_two_flags = []
sex_eth_group = []

# iterate through rows
for index in range(0, data.shape[0]):
    # initialize flags to 0
    g1_flag = 0
    g2_flag = 0

    # row values
    sex_val = data['sex'][index]
    ethnicity_binary_val = data['ethnicity_binary'][index]
    age_val = data['age'][index]
    high_chol_val = data['high_chol'][index]
    total_chol_val = data['total_chol'][index]
    has_hype_med_val = data['has_hype_med'][index]
    sbp_val = data['sbp'][index]
    is_diabetic_val = data['is_diabetic'][index]
    does_smoke_val = data['does_smoke'][index]

    # ethnicity = other
    if ethnicity_binary_val == 'Other':
        if sex_val == 'Female':
            ln_risk = (-29.799 * math.log(age_val) -
                       13.578 * math.log(high_chol_val) +
                       13.54 * math.log(total_chol_val) +
                       4.884 * math.log(age_val)**2 - 3.114 * math.log(age_val) * math.log(total_chol_val) +
                       3.149 * math.log(age_val) * math.log(high_chol_val) +
                       2.019 * (has_hype_med_val) * math.log(sbp_val) +
                       1.957 * (1 - has_hype_med_val) * math.log(sbp_val) +
                       0.661 * is_diabetic_val + 7.5740 * does_smoke_val +
                       -1.665 * does_smoke_val * math.log(age_val))
            risk_score = 1 - 0.9665 ** math.exp(ln_risk + 29.1817)
            group = 'Other Female'
        # other males
    else:
        ln_risk = (12.344 * math.log(age_val) +
```

Risk Score Calculation

```
# black males
else:
    ln_risk = (2.469 * math.log(age_val) +
               -0.307 * math.log(high_chol_val) +
               0.302 * math.log(total_chol_val) +
               1.916 * has_hype_med_val * math.log(sbp_val) +
               1.809 * (1 - has_hype_med_val) * math.log(sbp_val) +
               0.645 * is_diabetic_val +
               0.549 * does_smoke_val)
    risk_score = 1 - 0.8954 ** math.exp(ln_risk - 19.54)
    group = 'Black Male'

risk_scores.append(risk_score)
sex_eth_group.append(group)

# see if risk score passes threshold levels
if risk_score >= 0.075:
    g1_flag = 1

    if risk_score >= 0.100:
        g2_flag = 1

    group_one_flags.append(g1_flag)
    group_two_flags.append(g2_flag)

# add variable to dataframe
data['risk_score'] = risk_scores
data['g1_is_rec'] = group_one_flags
data['g2_is_rec'] = group_two_flags
data['group'] = sex_eth_group
```

EDA

```
table_data = all_data

# change labels for recommendation
table_data['g1_is_rec'] = np.where(table_data['g1_is_rec'] == '0', 'No', 'Yes')

# delete unnecessary columns
cols = table_data.columns.to_list()
indices = [16, 15, 13, 12, 0]

for index in indices:
    del cols[index]

# variables for table 1 creation
cat_vars = ['has_hype_med', 'sex', 'ethnicity', 'is_diabetic', 'does_smoke', 'has_hbp']
outcome_var = ['g1_is_rec']
labels = {'high_chol': 'high cholesterol',
          'total_chol': 'total cholesterol',
          'has_hype_med': 'is taking hypertension medication',
          'is_diabetic': 'is diabetic',
          'does_smoke': 'does smoke',
          'has_hbp': 'has high blood pressure',
          'g1_is_rec': 'Statin Recommendation'}

mytable = TableOne(table_data,
                   columns = cols,
                   categorical = cat_vars,
                   rename = labels,
                   groupby = outcome_var)

mytable
```

EDA

		Grouped by Statin Recommendation			
		Missing	Overall	No	Yes
n			13338	6771	6567
age, mean (SD)		0	57.5 (10.6)	50.5 (7.3)	64.8 (8.4)
high cholesterol, mean (SD)		0	53.6 (16.7)	55.9 (16.8)	51.2 (16.3)
total cholesterol, mean (SD)		0	194.6 (42.7)	196.7 (38.4)	192.4 (46.6)
is taking hypertension medication, n (%)		0	0	7596 (57.0)	5057 (74.7)
sex, n (%)	1		5742 (43.0)	1714 (25.3)	4028 (61.3)
	Female	0	6796 (51.0)	4392 (64.9)	2404 (36.6)
	Male		6542 (49.0)	2379 (35.1)	4163 (63.4)
	ethnicity, n (%)	0	1728 (13.0)	985 (14.5)	743 (11.3)
			1584 (11.9)	1008 (14.9)	576 (8.8)
			3274 (24.5)	1284 (19.0)	1990 (30.3)
			4829 (36.2)	2484 (36.7)	2345 (35.7)
			1508 (11.3)	792 (11.7)	716 (10.9)
			415 (3.1)	218 (3.2)	197 (3.0)
	is diabetic, n (%)	0	0	10773 (80.8)	6263 (92.5)
	1		2565 (19.2)	508 (7.5)	2057 (31.3)
	does smoke, n (%)	0	0	10793 (80.9)	5892 (87.0)
	1		2545 (19.1)	879 (13.0)	1666 (25.4)
sbp, mean (SD)		0	127.6 (18.3)	120.4 (14.4)	135.0 (19.0)
dbp, mean (SD)		0	73.2 (12.4)	73.7 (10.7)	72.7 (14.0)
has high blood pressure, n (%)		0	0	9219 (69.1)	4919 (72.6)
		1		4119 (30.9)	1852 (27.4)
				2267 (34.5)	

Analysis

```
# create plots of recommendation percentages
def create_rec_perc_plot(data, group_string, is_hbp):
    # subset data
    data_subset = data[data['group'] == group_string]

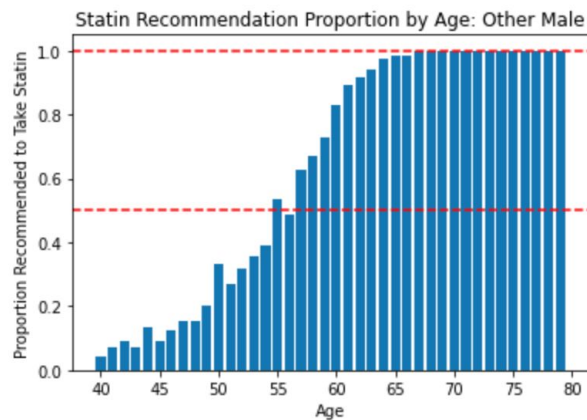
    # get recommendation percentages by age
    if is_hbp == 0:
        group_data = data_subset.groupby(['age'])['g1_is_rec'].value_counts(normalize=True).reset_index(name='proportion')
        rec_data = group_data[group_data['g1_is_rec'] == '1']
        rec_data = rec_data.drop('g1_is_rec', axis = 1)
    else:
        group_data = data_subset.groupby(['age'])['g2_is_rec'].value_counts(normalize=True).reset_index(name='proportion')
        rec_data = group_data[group_data['g2_is_rec'] == '1']
        rec_data = rec_data.drop('g2_is_rec', axis = 1)

    rec_data = rec_data.reset_index().drop('index', axis = 1)

    # create plot
    plt.bar(rec_data['age'], rec_data['proportion'])
    plt.axhline(y = 0.5, color = 'red', linestyle = '--')
    plt.axhline(y = 1.0, color = 'red', linestyle = '--')
    plt.title('Statin Recommendation Proportion by Age: {}'.format(group_string))
    plt.xlabel('Age')
    plt.ylabel('Proportion Recommended to Take Statin')
```

Analysis

```
create_rec_perc_plot(all_data, "Other Male", 0)
```



	age	proportion
--	-----	------------

0	40	0.040541
---	----	----------

1	41	0.072848
---	----	----------

2	42	0.089431
---	----	----------

3	43	0.070866
---	----	----------

4	44	0.136364
---	----	----------

	age	proportion
--	-----	------------

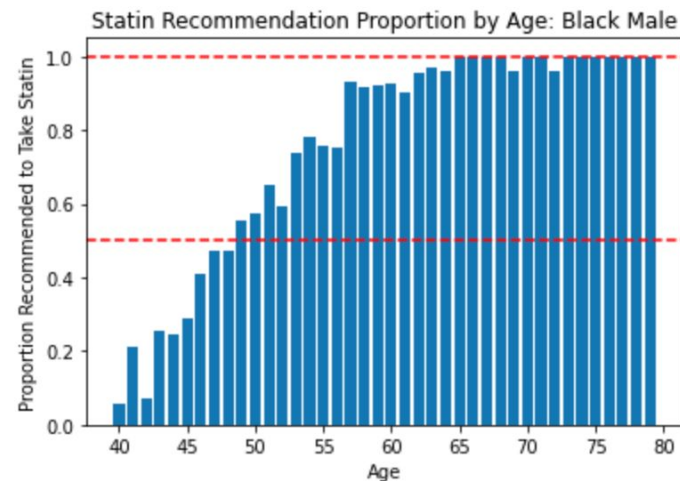
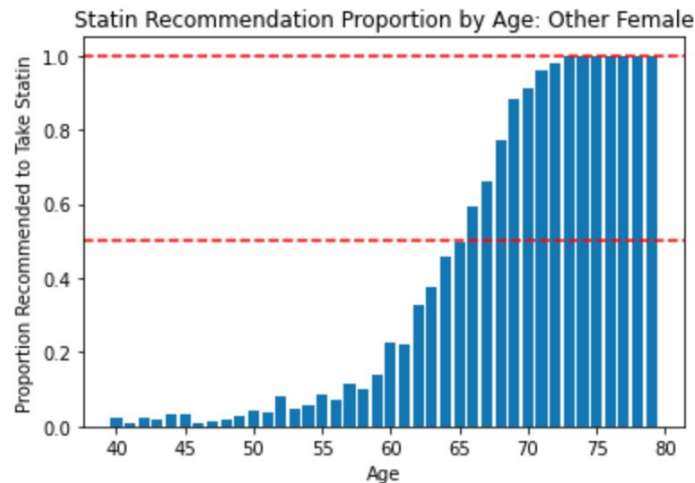
35	75	1.0
----	----	-----

36	76	1.0
----	----	-----

37	77	1.0
----	----	-----

38	78	1.0
----	----	-----

39	79	1.0
----	----	-----



Analysis

```
# get odds ratios for all data

# create dummy variables
all_data['is_female'] = np.where(all_data['sex'] == 'Female', 1, 0)
all_data['is_black'] = np.where(all_data['ethnicity_binary'] == 'Black', 1, 0)

# create datasets
x_data = all_data[['age', 'high_chol', 'total_chol', 'has_hype_med', 'is_female',
                    'is_black', 'is_diabetic', 'does_smoke', 'sbp', 'dbp']]
y_data = all_data['gl_is_rec']

x_data.apply(pd.to_numeric)
y_data.apply(pd.to_numeric)

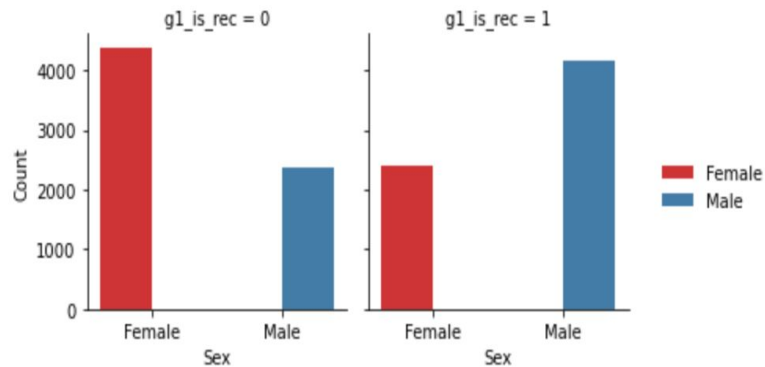
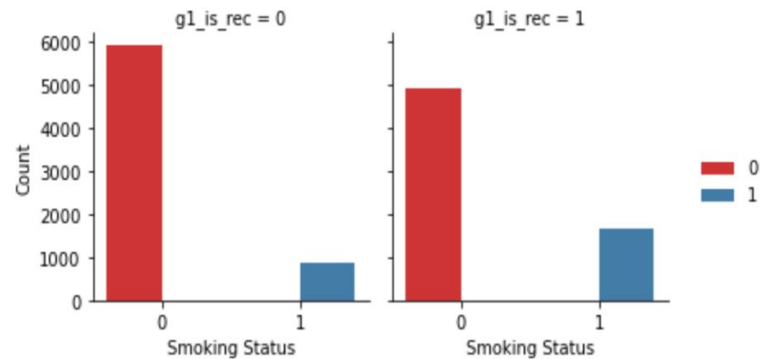
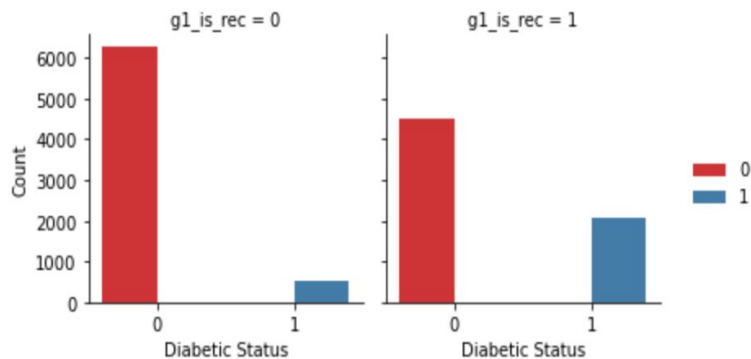
param_est = LogisticRegression().fit(x_data, y_data).coef_[0]
odds_ratios = []

for est in param_est:
    odds_ratios.append(math.exp(est))

cols = x_data.columns.tolist()
res = pd.DataFrame(cols)
res.columns = ['variable']
res['estimates'] = param_est
res['odds_ratio'] = odds_ratios
res
```

	variable	estimates	odds_ratio
0	age	0.455966	1.577697
1	high_chol	-0.078431	0.924566
2	total_chol	0.027603	1.027988
3	has_hype_med	2.446695	11.550113
4	is_female	-3.647265	0.026062
5	is_black	2.283844	9.814331
6	is_diabetic	6.153700	470.454639
7	does_smoke	3.815766	45.411514
8	sbp	0.081686	1.085115
9	dbp	-0.010388	0.989665

Analysis



Analysis

```
# get odds ratios for hbp data

# create dummy variables
hbp_data['is_female'] = np.where(hbp_data['sex'] == 'Female', 1, 0)
hbp_data['is_black'] = np.where(hbp_data['ethnicity_binary'] == 'Black', 1, 0)

# create datasets
x_data = hbp_data[['age', 'high_chol', 'total_chol', 'has_hype_med', 'is_female',
                    'is_black', 'is_diabetic', 'does_smoke', 'sbp', 'dbp']]
y_data = hbp_data['g1_is_rec']

x_data.apply(pd.to_numeric)
y_data.apply(pd.to_numeric)

param_est = LogisticRegression().fit(x_data, y_data).coef_[0]
odds_ratios = []

for est in param_est:
    odds_ratios.append(math.exp(est))

cols = x_data.columns.tolist()
res = pd.DataFrame(cols)
res.columns = ['variable']
res['estimates'] = param_est
res['odds_ratio'] = odds_ratios
res
```

	variable	estimates	odds_ratio
0	age	0.280848	1.324253
1	high_chol	-0.072359	0.930197
2	total_chol	0.006521	1.006542
3	has_hype_med	1.430246	4.179726
4	is_female	-2.564924	0.076925
5	is_black	1.553279	4.726945
6	is_diabetic	2.502132	12.208496
7	does_smoke	2.995653	19.998410
8	sbp	-0.006304	0.993716
9	dbp	-0.133272	0.875227

Analysis - Shapley Values

- Contribution of feature values to the difference between the actual and mean predicted value
- Identify variable influence in patients' 10-year risk prediction

```
# create dummy variables
all_data['is_female'] = np.where(all_data['sex'] == 'Female', 1, 0)
all_data['is_black'] = np.where(all_data['ethnicity_binary'] == 'Black', 1, 0)

# create datasets
x_data = all_data[['age', 'high_chol', 'total_chol', 'has_hype_med', 'is_female',
                  'is_black', 'is_diabetic', 'does_smoke', 'sbp', 'dbp']]
y_data = all_data['g1_is_rec']

x_data = x_data.apply(pd.to_numeric)
y_data = y_data.apply(pd.to_numeric)

x_data.head()
```

	age	high_chol	total_chol	has_hype_med	is_female	is_black	is_diabetic	does_smoke	sbp	dbp
0	44	28	190	0	1	0	0	0	118.000000	60.000000
1	43	73	169	0	1	1	0	1	102.000000	71.333333
2	51	43	198	0	0	0	0	1	146.666667	70.000000
3	55	54	206	0	0	0	0	0	124.666667	76.000000
4	70	35	118	0	0	1	0	0	158.666667	62.666667

Shapley - Subgroup

```
# by s+g
def create_shap_plot(data, group_string):
    # subset data
    data_subset = data[data['group'] == group_string]

    # create dummy variables
    data_subset['is_female'] = np.where(data_subset['sex'] == 'Female', 1, 0)
    data_subset['is_black'] = np.where(data_subset['ethnicity_binary'] == 'Black', 1, 0)

    # create datasets
    x_data = data_subset[['age', 'high_chol', 'total_chol', 'has_hype_med',
                          'is_diabetic', 'does_smoke', 'sbp', 'dbp']]
    y_data = data_subset['g1_is_rec']

    x_data = x_data.apply(pd.to_numeric)
    y_data = y_data.apply(pd.to_numeric)

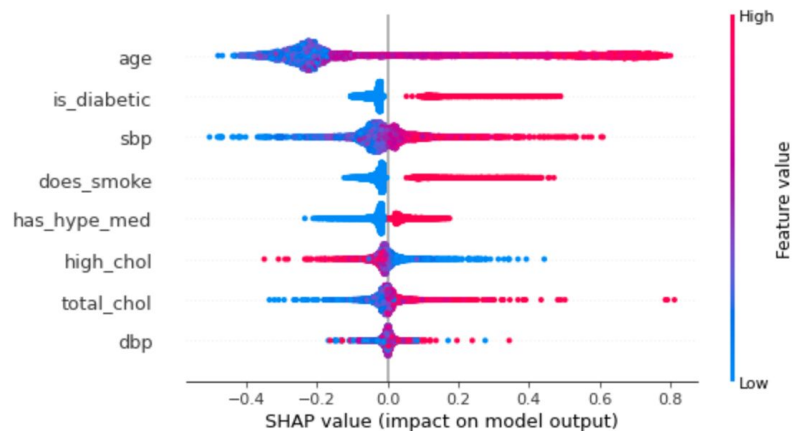
    # XGboost model
    mod = xgboost.XGBRegressor().fit(x_data, y_data)

    # get shapley values
    explainer = shap.Explainer(mod)
    shap_values = explainer(x_data)

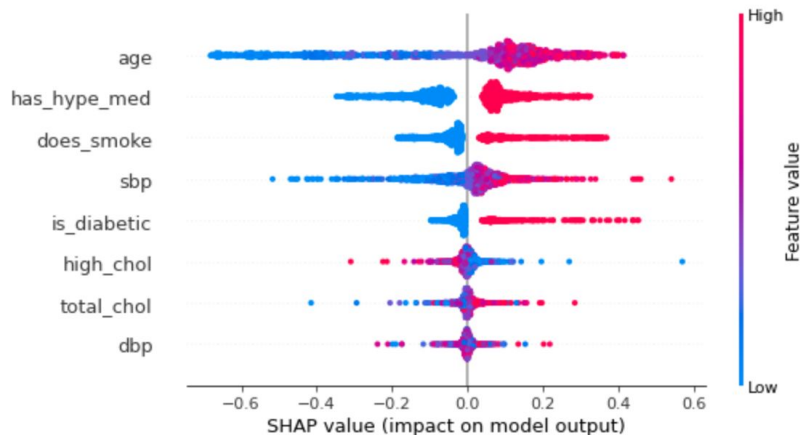
    # visualize
    shap.plots.beeswarm(shap_values)
```

Shapley - Subgroup

```
create_shap_plot(all_data, 'Other Female')
```

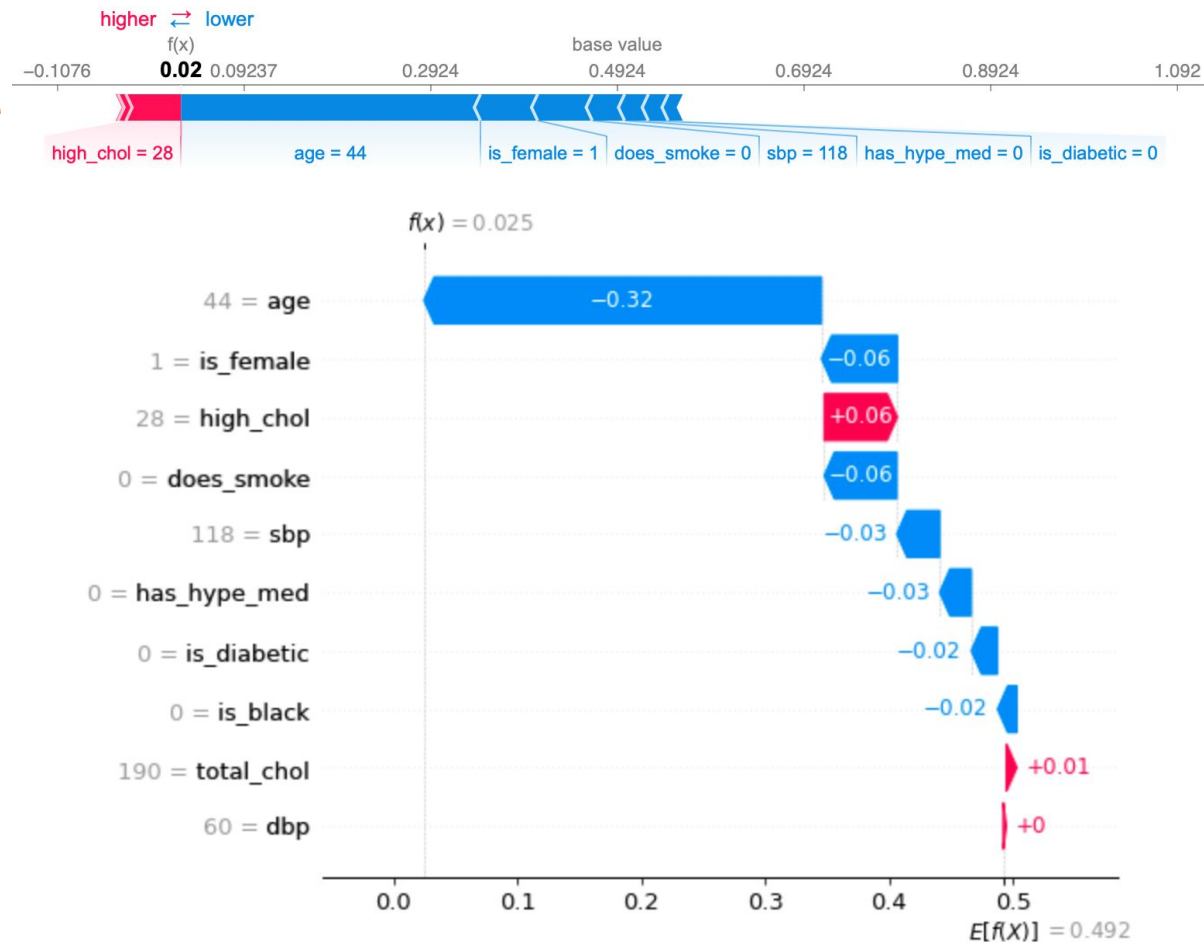


```
create_shap_plot(all_data, 'Black Male')
```



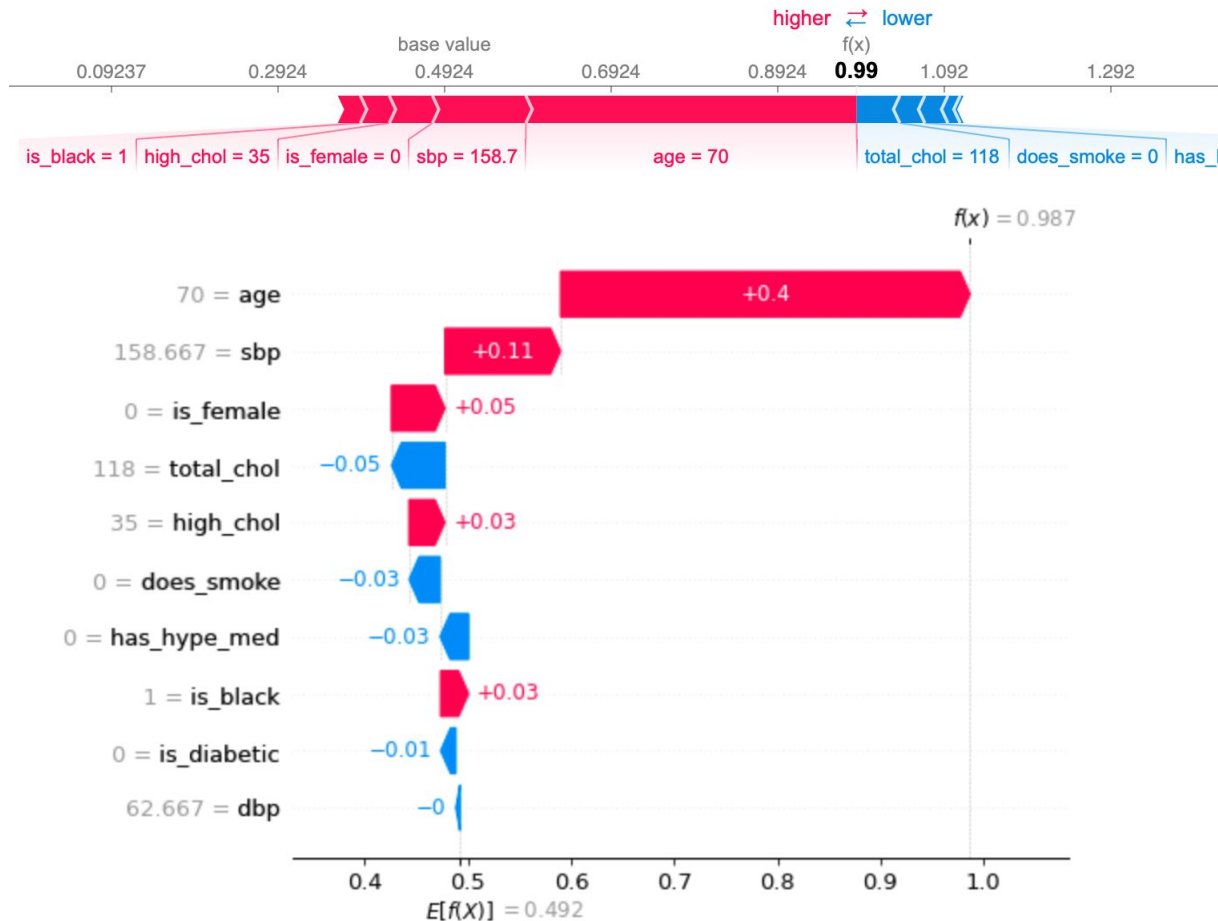
Shapley - Individual

	0
SEQN	62164
age	44
high_chol	28
total_chol	190
has_hype_med	0
sex	Female
ethnicity	Non-Hispanic White
is_diabetic	0
does_smoke	0
sbp	118.0
dbp	60.0
has_hbp	0
ethnicity_binary	Other
risk_score	0.017973
g1_is_rec	0
g2_is_rec	0
group	Other Female
is_female	1
is_black	0



Shapley - Individual

	4
SEQN	62191
age	70
high_chol	35
total_chol	118
has_hype_med	0
sex	Male
ethnicity	Non-Hispanic Black
is_diabetic	0
does_smoke	0
sbp	158.666667
dbp	62.666667
has_hbp	0
ethnicity_binary	Black
risk_score	0.16124
g1_is_rec	1
g2_is_rec	1
group	Black Male
is_female	0
is_black	1



Shapley - R Stuff

- Can't implement within Python due to package limitations
- Begins as before - get Shapley values
- Sort data by risk score
- Find index where recommendation flips

```
# prep data to sort
prepped_data <- data_numeric |> mutate(ID = 1:n()) |>
  rename(sort_var = risk_score) |> select(ID, sort_var)

# merge by ID and sort by risk score
data_merged <- merge(prepped_force, prepped_data, by = "ID")
sorted_data <- data_merged[order(sort_var),]

# remove variables so they aren't included in force plot
filtered_data <- sorted_data |> select(-sorted_id, -sort_var)

# create ID variable so plots are in sorted order
reset_index_data <- filtered_data |> mutate(ID = 1:n())

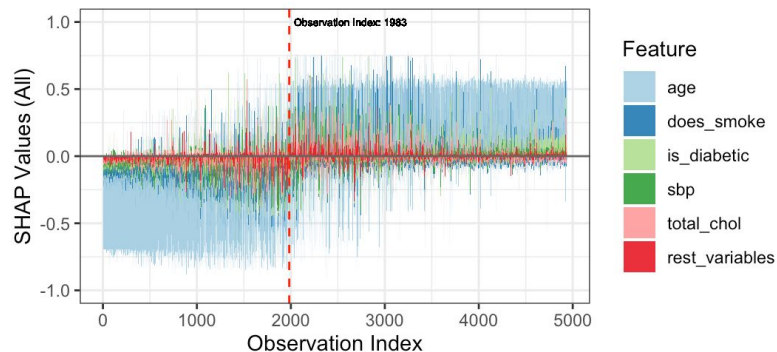
# get index cutoff for over/under threshold
temp <- data_numeric[order(data_numeric$risk_score),]
if (is_hbp_data == 0) {
  index_val <- min(which(temp$risk_score >= 0.075))
}
else {
  index_val <- min(which(temp$risk_score >= 0.010))
}

plot <- shap.plot.force_plot(reset_index_data, id = "ID", zoom_in = FALSE) +
  xlab("Observation Index") + ylim(-1, 1) +
  geom_vline(xintercept = index_val, linetype="dashed", color = "red") +
  geom_text(x=index_val+50, y=1,
            label=paste0("Observation Index: ", index_val),
            size=1.85, hjust = 0)

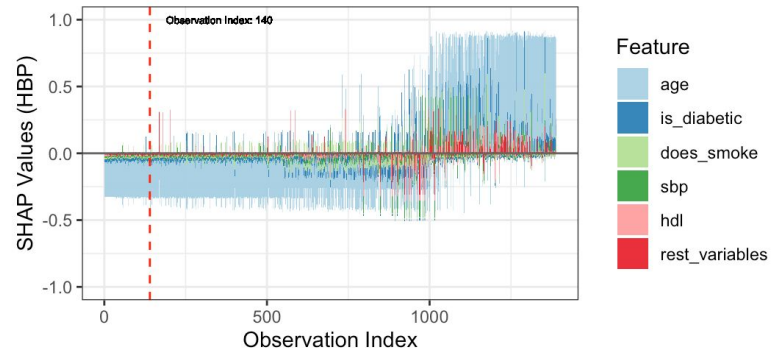
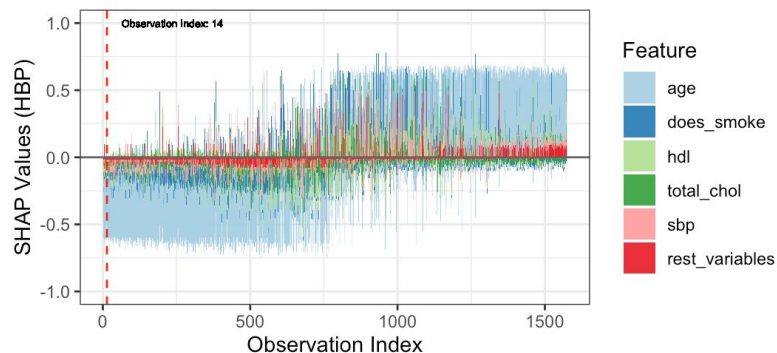
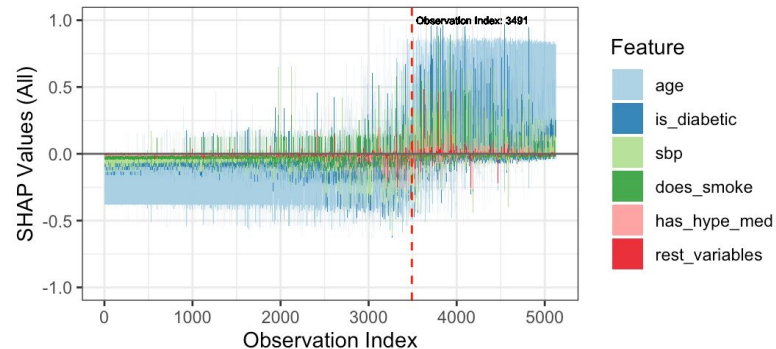
return(plot)
}
```

Shapley - R Stuff

Other Males by Risk Score



Other Females by Risk Score



Key Conclusions

- The most influential variables in predicting statin therapy recommendation are diabetes status, smoking status, hypertension medication status, and sex
- Recommendation rates from oldest to youngest:
 - Other Females → Black Females → Other Males → Black Males
- Black males' recommendation rates are much higher overall

Key Conclusions

- Ages when statin recommendation percentages initially pass 50% by subgroup
 - Black Females: 58
 - Black Males: 49
 - Other Females: 66
 - Other Males: 55
- Ages when statin recommendation percentages initially equal 100% by subgroup
 - Black Females: 68
 - Black Males: 65
 - Other Females: 73
 - Other Males: 67

Thank You!